इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY
सत्यमेव जयते

SwaYaan

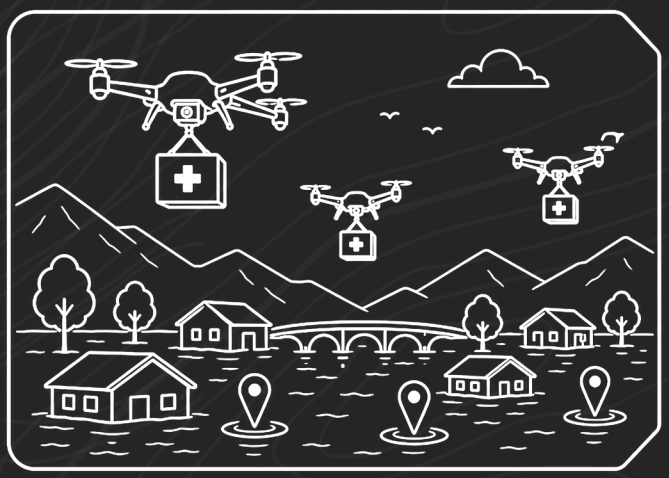
DRONE FEDERATION INDIA (DFI)

NIDAR

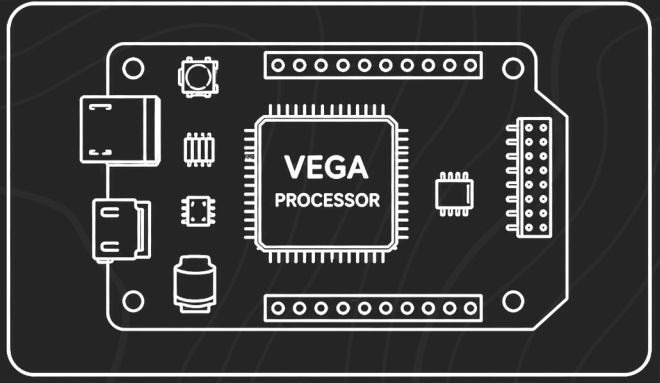
NATIONAL INNOVATION CHALLENGE FOR
DRONE APPLICATION AND RESEARCH

Building the Next Generation of Drone Innovators

RULEBOOK | 2026-27







www.nidar.org.in

Version: 2.1

About SwaYaan

SwaYaan, a national initiative by the Ministry of Electronics and Information Technology (MeitY), is dedicated to strengthening India's UAS/Drone ecosystem, aligning with the Government's vision to establish India as a global drone hub by 2030. The initiative empowers learners from undergraduates to faculty and opens learners across five key technical areas, conducting over 1,500 academic, research, and knowledge-sharing activities. Implemented through a network of 30 premier academic and R&D institutions, SwaYaan collaborates with skill councils and industry bodies to drive technological advancements, meet market demands, and set industry standards, ensuring a transformative impact on drone innovation.

About Drone Federation India

Drone Federation India (DFI) is a non-government, non-profit industry body representing over 550+ drone companies and 5,500+ drone pilots in the country, with a mission to promote the design, development, manufacturing, and export of Indigenous drone and counter-drone solutions in the country. DFI plays a pivotal role in policy advocacy, trade promotion, skill development, standards development, and community building by bringing together government, industry, academia, and civil society. DFI hosts several events, like Bharat Drone Mahotsav, which was inaugurated by the Hon'ble Prime Minister Shri Narendra Modi, and Bharat Drone Shakti, which was inaugurated by Hon'ble Defence Minister Shri Rajnath Singh. DFI drives India's leadership in the global landscape through its global alliances and collaborations.

About NIDAR

The National Innovation Challenge for Drone Application and Research (NIDAR) is India's largest and most impactful drone innovation challenge, designed to push the boundaries of engineering, innovation, and real-world problem-solving. It fosters cross-functional collaboration and focuses on addressing critical challenges in drone technology, particularly in disaster management. By engaging students with problem statements that reflect real-world needs, NIDAR encourages industry-oriented research, driving innovative solutions such as geotagging survivors and delivering survival kits using drones.

Beyond technical problem-solving, NIDAR provides participants with invaluable industry exposure through mentorship, guidance, and internship opportunities. This experience extends beyond engineering, equipping students with insights into business practices and entrepreneurship, helping them navigate the path to success in both technology and industry leadership. With its strong emphasis on applied research and industry collaboration, NIDAR aspires to develop the next generation of leaders in the drone sector, empowering them to drive the future of aerospace innovation.

Record of Revision

Date	Revision	Page Nos Affected
13 July 2026	Version 2 Launched	1 - 25
17 August 2026	Version 2.1 Published At the request of many participants: a) teams can be formed across multiple institutions. (3.2A) b) teams can be from one single branch (3.7A) To avoid confusion while referring to “drone missions”, the three main missions will be referred to as Problem Statements	 4 4 All

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1. About NIDAR

NIDAR 2025-26

The National Innovation Challenge for Drone Application and Research (NIDAR 2025-26), organised by the Drone Federation India in collaboration with the Ministry of Electronics and Information Technology under Project SwaYaan, emerged as one of India's largest student-focused drone innovation challenges. The previous edition was launched by the Secretary, MeitY, in the presence of the Chairman, AICTE, and leading industry stakeholders, and witnessed participation from 3,448 students across 22 States and 4 Union Territories, spanning 109 cities. The challenge focused on Disaster Management and Precision Agriculture. It successfully demonstrated the importance of hands-on innovation, indigenous capability development, and real-world problem-solving through drones and other emerging technologies.

Key Objectives of NIDAR

- To create a national innovation platform for student-led research and development in autonomous drone systems and indigenous avionics.
- To promote innovation in collaborative drone systems, GPS-denied navigation, indoor autonomy, and mission planning technologies.
- To encourage indigenous design and development of drone flight controllers, processors, and critical electronic subsystems.
- To strengthen collaboration between academia, government, defence stakeholders, and industry.
- To provide students with hands-on exposure to real-world operational and strategic drone challenges.
- To support India's vision of technological self-reliance and indigenous capability development in emerging technology sectors.

2. About the 2nd Edition

NIDAR 2026-27

Building on the success and national response to the first edition, NIDAR 2026-27 is envisioned as a larger national innovation platform to advance autonomous drone systems, indigenous avionics, autonomous navigation, and critical drone components. With the rapid growth of India's drone ecosystem and increasing strategic importance of autonomous systems, there is a strong need to encourage student-led innovation in areas such as GPS-denied navigation, collaborative drone operations, indigenous flight-control systems, and semiconductor-linked drone technologies. NIDAR 2026-27 seeks to bridge the gap between academia, industry, and government by providing students with opportunities to solve real-world operational challenges through applied innovation. This year's edition is divided into two tracks:

Track 1 - Drone Innovation

Track 1 focuses on the design, development and deployment of complete drone-based solutions for real-world challenges. Participants are required to integrate hardware, software, autonomy, sensing, communication and mission-planning capabilities to demonstrate a fully functional, end-to-end drone system. Track 1 has two problem statements.

Track 2 - Component Innovation

Track 2 focuses on the design, development and validation of critical drone subsystems and enabling technologies for unmanned aerial platforms. Participants are required to integrate hardware, software, sensors and control algorithms to develop and demonstrate a functional autopilot using the Aries v3 development board. Track 2 has 1 problem statement.

Problem Statement 1 - NIDAR RescueSwarm

A severe flash flood has affected a semi-urban settlement located near a river basin. Roads are blocked, several houses are inaccessible, and the local rescue team has received unverified reports of stranded people across a large area of the affected zone. Telecom towers in the affected region are down, mobile connectivity is unavailable, and ground teams cannot rely on external communication networks for real-time coordination. Ground access is delayed, and the first few minutes of response are critical for locating survivors and delivering basic medical support.

The rescue command centre has identified a 10-hectare search area where up to 10 survivors may be present. Since the area is too large for a single drone to cover efficiently, innovation teams must deploy a collaborative autonomous multi-drone system that can rapidly search the area, identify survivors, geotag their locations, and deliver a survivor kit to each identified survivor.

The goal of the innovation teams is to find survivors and deliver a survivor's kit to them as quickly as possible, while ensuring that all drones operate through a single ground control system, report mission data to a single operator, and complete the mission with minimal human intervention.

Problem Statement 2 - NIDAR AirMouse

A building has been severely affected by an earthquake or structural collapse, making several internal passages and rooms unsafe for immediate entry by rescue personnel due to debris, damaged structures, low visibility and the risk of further collapse. Initial reports indicate that survivors may be trapped inside, but their exact locations are unknown.

Since GPS signals are unavailable inside the building, rescue teams require an autonomous aerial system capable of entering the structure, navigating enclosed corridors and rooms, detecting survivors, generating a 2D map of the explored area and tagging survivor locations on the map.

Innovation teams shall develop an autonomous indoor drone system capable of entering a covered maze-like environment through a designated entry point, navigating passages and room-like sections, locating up to six survivors, generating a 2D map, tagging survivor locations and exiting through the designated exit point.

The objective is to autonomously navigate an unknown GPS-denied indoor environment and provide rescue teams with a usable 2D map showing the location of each detected survivor in the shortest possible time.

Problem Statement 3 - NIDAR VEGAPilot

India's drone ecosystem is expanding across defence, agriculture, logistics, infrastructure, disaster response, surveillance and public service delivery. However, the flight controller or autopilot remains one of the most critical drone subsystems, responsible for flight stability, navigation, communication, sensor integration and mission execution.

India's drone electronics ecosystem continues to depend significantly on imported flight controllers, processors, microcontrollers, integrated circuits and supporting components. Building a truly indigenous drone ecosystem, therefore, requires students and innovators to design, develop, integrate, and demonstrate reliable autopilot systems based on Indian processor platforms and non-Chinese electronic components.

Innovation teams shall develop a functional drone autopilot using the **Aeries v3 development board**. Each selected team shall receive two Aeries v3 development boards for development, testing and final integration. Teams shall design and develop the autopilot hardware and software stack around the board and demonstrate its ability to control a standard multirotor drone platform.

During the final demonstration, teams shall integrate their Aeries v3-based autopilot with a drone assembled using the standard parts list prescribed under this Rulebook. The platform may include an airframe, motors, ESCs, a GNSS module, a battery, a communication module, and other components.

The objective is to develop an indigenous Aeries v3-based autopilot capable of stabilising, controlling and navigating a real multirotor platform while demonstrating reliable integration with sensors, communication systems and mission-planning software.

Detailed Problem Statement Briefing

Detailed Briefings, including the objectives, arena specifications, technical requirements, operational conditions and specific rules for each problem statement, are provided in respective Annexures.

3. Team Eligibility

General Eligibility

3.1. A participant may be a member of only one team and may participate in only one problem statement.

3.2. Multiple teams from the same institution are permitted.

3.2A A team can consist of members from multiple institutions. A Letter of Approval can be provided by a single institution for all team members, or by the respective institutes of each team member.

3.3. No team shall share members, drones, flight controllers, processors, hardware, software, or competition deliverables with another team.

Track 1 – Drone Innovation

3.4. The target group shall consist of students enrolled in Diploma, Undergraduate, Postgraduate or Doctoral programmes in engineering, technology, science or related disciplines.

3.5. Students affiliated with research institutions, national laboratories, Centres of Excellence, government R&D organisations, or technical training institutions established, funded, operated, or accredited by the Government of India, any State Government, statutory body, university, or public sector undertaking shall also be eligible to participate, subject to verification by the Organising Team.

3.6. The student team can consist of a **minimum of 4 members and a maximum of 10 members.**

~~3.7. The student team **should be interdisciplinary** and have at least one member from a different branch of study or discipline to encourage diverse problem-solving approaches.~~

3.7A The student team may be interdisciplinary, from different branches of study or disciplines to encourage diverse problem-solving approaches.

3.8. Additionally, each team **should have one faculty member** in addition to the students for general guidance and mentorship.

3.9. All participants must remain enrolled with their respective institution or organisation until the conclusion of the **NIDAR Finals.**

Track 2 – Component Innovation

3.10. The target group shall consist of students enrolled in Diploma, Undergraduate, Postgraduate, or Doctoral programmes in engineering, technology, science, or related disciplines, with a focus on electronics, embedded systems, and related domains.

3.11. Students affiliated with research institutions, national laboratories, Centres of Excellence, government R&D organisations, or technical training institutions established, funded, operated, or accredited by the Government of India, any State Government, statutory body, university, or public sector undertaking shall also be eligible to participate, subject to verification by the Organising Team.

3.12. The student team can consist of a **minimum of 2 members and a maximum of 6 members.**

3.13. Additionally, each team should have **one faculty member** in addition to the students for general guidance and mentorship.

3.14. All participants must remain enrolled with their respective institution or organisation until the conclusion of the **NIDAR Finals as per the Competition Timeline in this Rulebook.**

4. Competition Structure

Phases		**PS 1 - NIDAR RescueSwarm**	**PS 2 - NIDAR RescueSwarm**	**PS 3 - NIDAR VegaPilot**
1	Application & Registration	Yes	Yes	Yes
2A	Verification Process	Yes	Yes	Yes
2B	Online Presentation & Shortlisting	**All Teams Auto-Shortlisted**		**Top 100**
Progress Review Rounds (Online)				
3A	Online Progress Review Round 1	Yes	Yes	Yes
3B	Online Progress Review Round 2	Yes	Yes	Yes
Final Mission (Offline)				
4A	Design Review Presentation	Yes	Yes	Yes
4B	Business Strategy Presentation	Yes	Yes	Yes
4C	Pre-Flight Inspection	Yes	Yes	Yes
4D	Final Mission (Flight)	Yes	Yes	Yes
NIDAR Awards				
5	Award & Felicitation Ceremony	Yes	Yes	Yes

Phase 1 - Application & Registration Fees

4.1. Each team must register online via the official competition website - www.nidar.org.in.

4.2. The registration deadline for the competition shall be posted on the official website.

4.3. In the event of an extension of the registration deadline, the same shall be updated on the official website and promptly communicated to all registered members.

4.4. Each team can choose only one mission in their online application form.

4.5. Each team shall submit the following documents in their online application:
- Letter of Approval from Academic Institution (Can be submitted later after registering online)
- Team Details (along with Govt ID Proof & College ID)
- Proof of Payment of Registration Fee

4.6. The Registration Fee for each team shall be INR 5,000 and shall be paid via the online portal.

4.7. Payment shall not be accepted in Cheque, Demand Draft or any other form.

4.8. The Registration Fee will be non-refundable in any case.

Phase 2A - Verification Process

4.9. The organising team shall verify each registration submission to confirm the completeness and adequacy of the information provided by the participating team.

4.10. Every complete application shall be shortlisted without the application of any additional evaluation or selection criteria. A confirmation email shall be sent to each shortlisted team.

4.11. The confirmation email shall constitute the official confirmation of selection for participation in the competition. The confirmation status shall also be reflected on the official website.

Phase 2B - Online Presentation & Shortlisting

4.12. Teams participating in Problem Statement 1 – NIDAR RescueSwarm and Problem Statement 2 – NIDAR AirMouse shall not be subject to any concept proposal shortlisting process. All teams that complete the registration requirements and submit the prescribed documents shall be automatically shortlisted and eligible to participate in the competition.

4.13. Teams participating in Problem Statement 3 – NIDAR VEGAPilot shall submit a Concept Proposal and undergo an online technical evaluation. Based on the evaluation, the top 100 teams shall receive two VEGA processors each and advance to the development stage.

Phase 3A & 3B - Progress Review Round 1 & 2

4.14. Progress Review Rounds 1 and 2 shall be conducted online to assess the development progress of participating teams. Attendance in both rounds shall be mandatory.

4.15. During each round, teams shall present updates on their development status, key milestones achieved, challenges encountered and planned next steps.

4.16. These rounds are intended solely for progress review and guidance. **No team shall be eliminated based on its performance in either round.** Where required, teams may receive feedback and guidance from the organising team, industry experts and mentors.

Phase 4A - Design Review Presentation

4.17. Each team shall deliver a comprehensive design review presentation to the jury in accordance with the prescribed schedule.

4.18. A total of 25 minutes shall be allocated to each team, comprising up to 15 minutes for the presentation and the remainder for questions and discussion.

4.19. The presentation shall, at a minimum, cover the topics specified under **Scoring Structure of this Rulebook**. Teams may also include any additional information they consider relevant to demonstrate their approach to system design, development, integration, and mission execution.

4.20. There shall be no restriction on the number of slides. However, the prescribed presentation time shall be strictly observed.

4.21. The detailed scoring criteria for the Design Review Presentation are provided in the subsequent sections of this Rulebook.

Phase 4B - Business Strategy Presentation

4.22. Each team shall deliver a comprehensive business strategy presentation to the jury in accordance with the prescribed schedule. The presentation shall be structured as a startup pitch, with the team presenting its drone system or solution as a viable business proposition.

4.23. A total of 25 minutes shall be allocated to each team, comprising up to 15 minutes for the presentation and the remainder for questions and discussion.

4.24. The presentation shall, at a minimum, cover the topics specified under the **Scoring Structure of this Rulebook.**

4.25. Each team shall also submit a detailed cost sheet setting out all costs and expenses incurred in designing and developing the complete solution for the competition.

4.26. There shall be no restriction on the number of slides. However, the prescribed presentation time shall be strictly observed.

4.27. Teams may include additional information they consider relevant to demonstrate their approach to market positioning, commercialisation, fundraising, scalability, and future business opportunities.

4.28. The detailed scoring criteria for the Business Strategy Presentation are provided in the subsequent sections of this Rulebook.

Phase 4C - Pre-Flight Inspection

4.29. Each team shall undergo a mandatory Pre-Flight Inspection, conducted by a designated jury member serving as the Flight Inspector, before undertaking the Final Mission.

4.30. The inspection shall verify the safety, readiness and compliance of the drone system, Command and Control Station and associated equipment with the applicable technical and operational requirements.

4.31. A model Pre-Flight Inspection Checklist shall be released separately.

4.32. Only teams that successfully pass the Pre-Flight Inspection shall be permitted to proceed to the Final Mission.

Phase 4D - Final Mission

4.33. Each team shall be permitted to undertake the Final Mission only after successfully completing the Pre-Flight Inspection.

4.34. The team shall position its Command and Control Station, including all associated equipment such as antennas, displays, computers and remote-control devices, within the designated area. A maximum of two team members shall be permitted to operate or supervise the Command and Control Station.

4.35. Each drone shall be positioned within the designated launch or staging area. One team member may be assigned to position and supervise each drone. The same team member may reset, reload or reposition the assigned drone during the mission, where permitted under the relevant mission rules.

4.36. Each team shall be provided a maximum setup time of five minutes. If the team fails to complete the setup within this period, the Final Mission timer may be started at the jury's discretion.

4.37. For **Problem Statement 1 – NIDAR RescueSwarm**, teams shall demonstrate autonomous multi-drone collaboration for survivor detection, geotagging, mission coordination and survival-kit delivery within the designated mission area.

4.38. For **Problem Statement 2 – NIDAR AirMouse**, teams shall demonstrate autonomous navigation within a GPS-denied indoor environment, survivor detection and localisation, map generation, mission reporting and successful exit from the designated mission area.

4.39. For **Problem Statement 3 – NIDAR VEGAPilot**, teams shall demonstrate the functionality of their VEGA processor-based autopilot through integration with the prescribed drone platform, mission execution, telemetry communication, mission planner integration, and safe flight operations.

4.40. The detailed Mission Briefing, including mission-specific objectives, scoring criteria, technical specifications, arena requirements and operational conditions, is provided in the respective Annexure to this Rulebook.

4.41. The maximum time permitted for each Final Mission shall be 30 minutes. Any objective completed or points earned after the expiry of the prescribed time limit shall not be considered for scoring.

Phase 5 - NIDAR Awards

4.42. The NIDAR Awards Ceremony shall be held at the conclusion of the competition to recognise outstanding performance across all problem statements.

4.43. Winning teams shall be presented with trophies, awards, certificates and other forms of recognition based on the applicable scoring criteria and their overall performance in the competition.

4.44. The organising team reserves the right to introduce additional awards, special recognitions, or category-specific honours as deemed appropriate.

5. Competition Timeline

5.1. All participating teams must note the essential deadlines as stated in this section. Failing to comply with any deadline may result in disqualification from the competition. Organisers shall have the right to extend deadlines for broader participation and accommodate students' academic activities.

5.2. Important deadlines concerning the competition are as follows:

Phases		**Tentative Timeline**
1	Application & Registration Deadline	2nd Week of August 2026
2A	Verification Process	3rd Week of August 2026
2B	Online Presentation & Shortlisting	3rd Week of September 2026
Progress Review Rounds (Online)		
3A	Online Progress Review Round 1	2nd Week of October 2026
3B	Online Progress Review Round 2	2nd Week of December 2026
Final Mission (Offline)		
4A	Design Review Presentation	January 2027
4B	Business Strategy Presentation	
4C	Pre-Flight Inspection	
4D	Final Mission (Flight)	
NIDAR Awards		
5	Award & Felicitation Ceremony	January 2027

6. Awards & Recognition

No.	Award(s)	PS 1 - NIDAR RescueSwarm	PS 2 - NIDAR AirMouse	PS 3 - NIDAR VegaPilot
1	NIDAR Gold Award	INR 8,00,000	INR 8,00,000	INR 4,00,000
2	NIDAR Silver Award	INR 5,50,000	INR 5,50,000	INR 3,00,000
3	NIDAR Bronze Award	INR 4,00,000	INR 4,00,000	INR 2,00,000
4	NIDAR Ignite Prize (x5)	INR 1,00,000	INR 1,00,000	INR 1,00,000
5	NIDAR Design - Gold Award	INR 75,000	INR 75,000	INR 75,000
6	NIDAR Design - Silver Award	INR 50,000	INR 50,000	INR 50,000
7	NIDAR Business - Gold Award	INR 75,000	INR 75,000	INR 75,000
8	NIDAR Business - Silver Award	INR 50,000	INR 50,000	INR 50,000
9	NIDAR Diaries	INR 50,000		
10	NIDAR Participation Certificate	All participating teams and team members		
11	Additional Benefits (to be announced soon)	Incubation support, Cloud Credits, Design and Simulation Software, Internship Opportunities		

7. Deliverables

7.1. Participating teams shall submit the following deliverables at the relevant stages of the competition:

7.2. **Registration Documents:** Completed online application form, approval letter from the academic institution, team details, and proof of payment of the registration fee.

7.3. **Phase 2B – Concept Proposal and Shortlisting:** A comprehensive Concept Proposal presentation for **Problem Statement 3 – NIDAR VEGAPilot**, submitted in PDF or PPT/PPTX format.

7.4. **Phase 4A – Design Review Presentation:** A comprehensive Design Review presentation submitted in PDF or PPT/PPTX format.

7.5. **Phase 4B – Business Strategy Presentation:** A comprehensive Business Strategy presentation submitted in PDF or PPT/PPTX format; and

Bill of Materials and Cost Sheet: A detailed bill of materials and cost sheet specifying all costs and expenses incurred by the team in designing, developing, integrating and testing its solution.

8. General Technical and Operational Requirements

General Rules

8.1. Teams shall independently design, develop, integrate and demonstrate their complete solutions.

8.2. The use of commercially available, market-ready or ready-to-fly complete drone airframes shall not be permitted. Teams may use commercially available components, open-source software and custom-developed hardware or software, unless otherwise specified in the relevant Problem Statement Annexure.

8.3. All systems and operations shall comply with the applicable safety requirements, technical specifications and competition rules.

8.4. During mission execution, the use of GSM, LTE, 5G, public Wi-Fi, internet connectivity, cloud-based services or any other external network shall be prohibited.

8.5. All communication between drones, onboard systems and the Ground Control Station shall operate through locally deployed communication systems. Optical-fibre cables, wired communication links, physical tethers or any other cable connected to a drone during flight shall not be permitted.

8.6. The organising team and jury reserve the right to inspect any hardware, software, firmware, communication system, source configuration or component used by a participating team for the purpose of verifying compliance.

8.7. Unsafe operation, deliberate interference with another team or competition infrastructure, misrepresentation of system capabilities, or violation of the competition rules may result in penalties, suspension from the mission or disqualification.

Rules for PS1 - NIDAR RescueSwarm

8.8. Each team shall deploy at least two or more drones operating as a coordinated multi-drone mission system.

8.9. The combined all-up weight of all deployed drones, including batteries, payloads, sensors and communication equipment, shall not exceed 25 kg.

8.10. All drones shall take off from and land within the designated 12 ft × 12 ft launch and landing area.

8.11. The system shall autonomously search the designated mission area and detect up to ten survivors. Each detected survivor shall be autonomously geotagged, and the corresponding location shall be displayed on the Ground Control Station.

8.12. Teams shall autonomously deliver survival kits to detected survivors. Each survival kit shall weigh 200 grams and measure 20 cm × 10 cm × 5 cm. Payload deployment shall be performed autonomously.

8.13. All drones shall operate through a single Ground Control Station and a unified operator interface. Mission data from all drones shall be consolidated and displayed through the same interface.

8.14. The Ground Control Station shall display, at a minimum:
- Mission status
- Live camera feed from each drone
- Position or estimated position of each drone
- Assigned search area or task for each drone
- Detected and geotagged survivor locations
- Survival-kit delivery status
- Communication and system health status
- Consolidated mission progress and reporting

8.15. The system shall demonstrate collaborative mission execution, including area allocation, task distribution, mission coordination and consolidated mission reporting.

8.16. Any manual waypoint modification, flight-path correction, payload-release command, survivor tagging or mission replanning during mission execution shall be treated as manual intervention.

8.17. The system shall not rely on any external communication network for mission execution, inter-drone coordination or data exchange.

8.18. All drones shall remain within the mission boundary specified by the organising team.

8.19. Each drone shall incorporate, at a minimum, Return-to-Home, communication-loss recovery, low-battery fail-safe, geofence protection and mission-abort capabilities.

Rules for PS2 - NIDAR AirMouse

8.20. Each team shall deploy an autonomous drone capable of operating within a GPS-denied indoor environment.

8.21. The drone shall autonomously enter and exit the arena through the same designated access point.

8.22. The total maze or arena area shall not exceed 15 m × 15 m. Each corridor shall provide a minimum clear width of 1 m and a minimum vertical clearance of 8 ft. The standard room dimensions shall be 2 m × 2 m.

8.23. The drone shall take off from within the designated 2-ft × 2-ft launch area.

8.24. The drone shall autonomously navigate through corridors, rooms, turns, junctions and obstacles within the indoor arena. GPS- or GNSS-based navigation shall not be used during mission execution.

8.25. The system shall detect and localise up to six survivors positioned within the arena. Each detected survivor shall be autonomously tagged at the corresponding location on the generated map.

8.26. The system shall generate and display a real-time two-dimensional map of the explored environment.

8.27. The Ground Control Station shall display, at a minimum:
- Mission status
- Live camera feed from the drone
- Drone position or estimated position
- Generated map
- Tagged survivor locations
- Mission progress

8.28. Any manual navigation assistance, map editing, localisation correction, survivor tagging or path-planning input during mission execution shall be treated as manual intervention.

8.29. The drone shall be designed for safe operation within confined indoor spaces and shall avoid contact with walls, ceilings, arena structures and obstacles.

8.30. The drone shall be equipped with propeller covers or guards that fully protect the propeller operating area.

8.31. The drone shall incorporate, at a minimum, emergency-stop, mission-abort, communication-loss recovery and low-battery fail-safe capabilities.

Rules for PS3 - NIDAR VEGAPilot

8.32. Teams shall develop a functional flight controller and autopilot system using the Aeries v3 development board as the primary processing platform.

8.33. Participation in the development stage shall be limited to teams shortlisted through the Concept Stage. Each shortlisted team shall receive two Aeries v3 development boards for development, integration, testing and final demonstration.

8.34. Teams shall build and integrate a complete multirotor drone using the standard parts list prescribed under this Rulebook. No additional autopilot-related components shall be used unless expressly permitted by the organising team.

8.35. The autopilot shall support, at a minimum:
- Arming and disarming
- Stabilised flight
- Altitude hold
- Position hold
- Manual flight-mode control
- Safe landing

8.36. The autopilot shall interface with the prescribed sensors, GNSS module, electronic speed controllers, motors and communication systems required for flight operations.

8.37. The autopilot shall communicate with a compatible Mission Planner or Ground Control Station to support:
- Vehicle configuration
- Telemetry transmission

- Vehicle monitoring
- Flight-mode display
- Flight-data recording

8.38. The Ground Control Station shall display, at a minimum:
- Vehicle status
- Vehicle location
- Active flight mode
- Battery status
- Relevant flight and telemetry information

8.39. The system shall record flight and telemetry data for post-flight review.

8.40. During the final demonstration, teams shall integrate the Aeries v3-based autopilot with the drone platform assembled using the standard parts list prescribed under this Rulebook.

8.41. The demonstration shall include:
- Successful arming and take-off
- Taking off and landing in a designated launch and recovery area
- Stabilised flight
- Position hold
- Manual navigation through the prescribed flight course or obstacles
- Autonomous navigation through 5 waypoints
- Telemetry communication with the Ground Control Station
- Return to home
- Safe landing

8.42. All components incorporated within the autopilot system shall comply with the indigenous-component requirements prescribed by the organising team. Chinese-origin components shall not be used within the autopilot system.

8.43. Teams shall submit a complete Bill of Materials identifying the make, model, manufacturer, supplier and country of origin of each component used.

8.44. The autopilot shall incorporate fail-safe provisions for:
- Low-battery conditions
- Communication loss
- GNSS failure
- Unstable flight conditions
- Mission abort
- Emergency landing

8.45. Teams shall not modify, damage or permanently alter any standard component provided by the organising team unless expressly permitted.

8.46. The final demonstration shall establish the successful integration of the Aeries v3-based autopilot, stable flight performance, position-hold capability, telemetry communication, flight data logging, and safe operation of the assembled drone platform.

9. Scoring Structure

Overall Scoring Criteria

Final Mission		Maximum Points
4A	Design Review Presentation	200
4B	Business Strategy Presentation	200
4C	Pre-Flight Inspection	Pass or Fail
4D	Final Mission (Flight)	600
	Total	1000

Problem Statement 1 - NIDAR RescueSwarm

Phase 4A - Design Review Presentation		
No.	**Parameter & Description**	**Max Score**
1	Introduction, Team Composition & Domain Diversity	10
2	Overall Mission Architecture & System Design	20
3	Technical Approach, Drone Design & Indigenisation Strategy	20
4	Multi-Drone Collaboration & Task Allocation	30
5	Survivor Detection, Geotagging & Accuracy	30
6	Payload Delivery & Drop Accuracy	25
7	Autonomous Mission Execution & Time Efficiency	30
8	Ground Control Station, Mission Map & Multi-Drone Status Reporting	20
9	Fail-Safe Features & Safety	15
	TOTAL	**200**

Phase 4B - Business Strategy Presentation		
No.	**Parameter & Description**	**Max Score**
1	Problem Understanding & Real-World Relevance	30
2	Target Users, Customers & Beneficiaries	20
3	Market Sizing & Deployment Potential	30
4	Business Model & Revenue Approach	30
5	Expenditure Breakdown & Resource Planning	20
6	Funds Raised, Sponsorships & Resource Mobilisation	20
7	Competitive Advantage & Differentiation	20
8	Go-to-Market Strategy & Partnership	20
9	Regulatory, Safety & Adoption Readiness	10
	TOTAL	**200**

Phase 4C - Pre-Flight Inspection		
No.	**Parameter & Description**	**Result**
1	Pre-flight Inspection shall only be a deciding phase, and successful completion of this phase will allow team members to participate in the Final Mission. Teams shall only be permitted to retry this phase once.	Pass or Fail

Phase 4D - Final Mission - Scoring			
No.	**Criteria**	**Scoring Method**	**Max Score**
1	**Survivor Detection & Geotagging**	25 marks for each correctly detected survivor whose location is also correctly geotagged on the GCS / mission map. Maximum 10 survivors.	250
2	**Survivor Kit Delivery Accuracy**	Marks are awarded for each successful survivor kit drop based on the accuracy zone. Maximum 10 successful drops. Delivery in Zone A (within 1 metre): 20 per drop Delivery in Zone B (within 2 metres): 14 per drop Delivery in Zone C (within 3 metres): 8 per drop	200

3	**Multi-Drone Collaborative Mission Execution**	Yes/No. Full 50 marks if two or more drones operate as one coordinated mission system with shared mission execution, common mission logic, and no independent manual control. Zero if not achieved.	50
4	**Single GCS / Mission Planner Interface**	Yes/No. Full 50 marks if all drones, survivor tags, delivery status, mission boundary, and mission progress are visible in a single operator interface. Zero if not achieved.	50
5	**Fast Completion Bonus**	Yes/No. Full 50 marks if the mission is completed within half of the maximum permitted mission time. Zero if not achieved.	50
	TOTAL		**600**

Phase 4D - Final Mission - Penalties			
No.	**Penalty Condition**	**Penalty Description**	**Max Score**
1	**Landing outside the designated launch or landing zone**	For each drone that lands outside the designated landing zone after mission completion, except in an organiser-approved emergency landing.	-10 per drone
2	**Geofence / Mission Boundary Breach**	For each instance where any drone crosses outside the assigned mission boundary/geofence.	-20 per instance
3	**Repeated Geofence Breach by Same Drone**	If the same drone breaches the geofence more than once during the mission.	Additional -20 per drone
4	**Manual Input or Reset**	The operator provides manual navigation input, flight control input, waypoint change input, payload drop input, survivor tagging input, system reset, or mission reset after mission start, other than safety abort or emergency recall.	-50 per instance
5	**Crash**	For each instance where any drone crashes during the mission, including uncontrolled ground impact, collision resulting in loss of flight, or crash landing.	-50 per crash
	TOTAL		**600**
Total penalties may be capped at 150 marks, except for safety-critical violations where the organisers may terminate the mission or disqualify the team.			

Problem Statement 2 - NIDAR AirMouse

Phase 4A - Design Review Presentation		
No.	**Parameter & Description**	**Max Score**
1	Introduction, Team Composition & Domain Diversity	10
2	Overall Solution Architecture & Indoor Autonomy Strategy	25
3	Technical Approach, Drone Design & Indigenisation Strategy	20
4	GPS-Denied Navigation & Confined-Space Flight	30
5	Indoor Mapping Quality & Mission Planner Display	30
6	Survivor Detection, Localisation & Tagging Accuracy	35
7	Autonomous Mission Execution & Time Efficiency	20
8	Ground Control Station, Mission Map & Survivor Status Reporting	15
9	Fail-Safe Features & Safety	15
	TOTAL	**200**

Phase 4B - Business Strategy Presentation		
No.	**Parameter & Description**	**Max Score**
1	Problem Understanding & Real-World Relevance	30
2	Target Users, Customers & Beneficiaries	20
3	Market Sizing & Deployment Potential	30
4	Business Model & Revenue Approach	30
5	Expenditure Breakdown & Resource Planning	20
6	Funds Raised, Sponsorships & Resource Mobilisation	20
7	Competitive Advantage & Differentiation	20
8	Go-to-Market Strategy & Partnership	20
9	Regulatory, Safety & Adoption Readiness	10
	TOTAL	**200**

Phase 4C - Pre-Flight Inspection		
No.	**Parameter & Description**	**Result**
1	Pre-flight Inspection shall only be a deciding phase, and successful completion of this phase will allow team members to participate in the Final Mission. Teams shall only be permitted to retry this phase once.	Pass or Fail

Phase 4D - Final Mission - Scoring			
No.	**Criteria**	**Scoring Method**	**Max Score**
1	**Survivor Detection & Localisation on 2D Map**	40 marks for each correctly detected survivor whose location is also correctly tagged on the generated 2D map. Location tagging shall mean marking the survivor location with the correct grid reference on the 1 m x 1 m reference grid. Maximum 6 survivors.	240
2	**2D Grid Mapping Accuracy**	Marks awarded based on the number of correctly mapped 1 m x 1 m grid cells within the 15 m x 15 m grid zone, verified against the organiser's reference grid layout. For each grid cell, the team must correctly map the presence of walls, openings, corridors or accessible directions adjoining that grid cell.	220
3	**GCS / Mission Planner Map Display**	Yes/No. Full marks if the GCS displays the generated 2D map, estimated drone position, survivor tags and mission progress. Zero if not achieved.	50
4	**Autonomous Exit from Designated Exit Point**	Yes/No. Full marks if the drone exits from the designated exit point without manual intervention. Zero if not achieved.	50
5	**Fast Completion Bonus**	Yes/No. Full marks if the mission is completed within half of the maximum permitted mission time. Zero if not achieved.	25
6	**Safe Mission Completion**	Yes/No. Full marks if the mission is completed without crash, unsafe contact, uncontrolled landing, or organiser intervention. Zero if not achieved.	15
	TOTAL		**600**

Phase 4D - Final Mission - Penalties			
No.	**Penalty Condition**	**Penalty Description**	**Max Score**
1	**Landing outside the designated launch/landing zone**	For each drone that lands outside the designated landing zone after mission completion, except in an organiser-approved emergency landing.	-10 per drone
2	**Geofence / Mission Boundary Breach**	For each instance in which any drone crosses outside the assigned indoor mission boundary/geofence/permitted maze area.	-20 per instance
3	**Repeated Geofence Breach by Same Drone**	If the same drone breaches the geofence / permitted maze area more than once during the mission.	Additional -20 per drone
4	**Manual Input or Reset**	The operator provides manual navigation input, flight control input, path correction input, survivor tagging input, map correction input, and system or mission reset after mission start, other than safety abort or emergency recall.	-50 per instance
5	**Crash**	For each instance where any drone crashes during the mission, including uncontrolled ground impact, collision resulting in loss of flight, or crash landing.	-50 per crash
Total penalties may be capped at 150 marks, except for safety-critical violations, for which the organisers may terminate the mission or disqualify the team.			

Problem Statement 3 - NIDAR VEGAPilot

Phase 2B - Online Presentation & Shortlisting (Top 100 Team Selection)		
No.	**Parameter & Description**	**Max Score**
1	Introduction, Team Composition & Domain Diversity	15
2	Prior Experience & Demonstrated Capability	15
3	Understanding of the Problem Statement	15
4	Preliminary Technical Approach	20
5	Design Tools, Development Tools & Engineering Methods	10
6	Indigenous Component & Non-Chinese Supply Chain Approach	10

7	Development Roadmap, Timelines & Feasibility	15
	TOTAL	**100**

Phase 4A - Design Review Presentation		
No.	**Parameter & Description**	**Max Score**
1	Introduction, Team Composition & Domain Diversity	10
2	Overall Solution Architecture & Aries v3 Integration Strategy	20
3	Autopilot Board Design & Hardware Integration	20
4	Sensor Integration, Navigation & Flight Control	30
5	Software, Firmware & Flight-Control Stack	25
6	Mission Planner & Ground Control Station Integration	20
7	Flight Capability & System Demonstration Strategy	20
8	Indigenous Components, BOM & Supply Chain Compliance	20
9	Safety, Failsafe, Testing & Validation Strategy	35
	TOTAL	**200**

Phase 4B - Business Strategy Presentation		
No.	**Parameter & Description**	**Max Score**
1	Problem Understanding & Real-World Relevance	30
2	Target Users, Customers & Beneficiaries	20
3	Market Sizing & Deployment Potential	30
4	Business Model & Revenue Approach	30
5	Expenditure Breakdown & Resource Planning	20
6	Funds Raised, Sponsorships & Resource Mobilisation	20
7	Competitive Advantage & Differentiation	20
8	Go-to-Market Strategy & Partnership	20

9	Regulatory, Safety & Adoption Readiness	10
	TOTAL	**200**

Phase 4C - Pre-Flight Inspection		
No.	**Parameter & Description**	**Result**
1	Pre-flight Inspection shall only be a deciding phase, and successful completion of this phase will allow team members to participate in the Final Mission. Teams shall only be permitted to retry this phase once.	Pass or Fail

Phase 4D - Final Mission - Scoring			
No.	**Criteria**	**Scoring Method**	**Max Score**
1	**Manual Flight and Flight Stabilisation**	a. Controlled flight response: 30 marks each for successfully demonstrating: (i) controlled roll and pitch response; and (ii) controlled yaw response. b. Stable hover: 20 marks for maintaining a stable hover for a minimum of 30 seconds.	80
2	**Assisted Flight Modes**	50 marks each for successfully demonstrating: (i) Position Hold within ±3 m for a minimum of 30 seconds; and (ii) Altitude Hold within ±1 m for a minimum of 30 seconds.	100
3	**Autonomous Waypoint Mission**	40 marks for each organiser-defined waypoint, up to a maximum of five waypoints, successfully reached and held for a minimum of five seconds before the aircraft automatically proceeds to the next waypoint. The mission shall be executed without manual intervention after initiation. Any manual intervention shall terminate scoring for all remaining waypoints.	200
4	**Return-to-Launch (RTL)**	120 marks if the autopilot successfully initiates Return-to-Launch and completes an autonomous landing. 80 marks if Return-to-Launch is successfully completed, but manual intervention is required only for the final landing due to safety considerations. No marks shall be awarded if Return-to-Launch is not successfully completed.	120
5	**Mission Planner / Ground Control Station Integration**	20 marks each for successfully demonstrating: (i) vehicle connection; (ii) mission upload; (iii) live telemetry; (iv) battery-status display; and (v) mission-progress display.	100
	TOTAL		**600**

Phase 4D - Final Mission - Penalties			
No.	**Penalty Condition**	**Penalty Description**	**Max Score**
1	**Position Hold exceeds ±3 m tolerance**	For each instance in which the aircraft fails to remain within ±3 m of the commanded position during the required 30-second hold period.	−10 per instance
2	**Altitude Hold exceeds ±1 m tolerance**	For each instance in which the aircraft fails to remain within ±1 m of the commanded altitude during the required 30-second hold period.	−10 per instance
3	**Landing outside the designated landing area**	Where the aircraft completes the demonstration but comes to rest outside the organiser-designated landing area, except in an organiser-approved emergency landing.	−20 per instance
4	**Obstacle Contact**	For each instance in which the aircraft, propeller guard, or any attached component contacts an obstacle or course structure.	−10 per instance
5	**Course Boundary Breach**	For each instance in which the aircraft crosses the organiser-defined flight-course boundary.	−20 per instance
6	**Organiser Intervention**	Where the organiser or Flight Inspector is required to intervene to prevent an unsafe condition or recover the aircraft.	−50 per instance
7	**Crash or Uncontrolled Landing**	For each crash, uncontrolled ground impact, collision resulting in loss of controlled flight or crash landing.	−50 per instance
Total penalties may be capped at 150 marks, except in cases involving unsafe operation, deliberate rule violations or safety-critical incidents, for which the organising team may terminate the demonstration or disqualify the team.			

Tie Breaking

9.1. For awards based solely on the Design Review or Business Strategy, , the team with the higher Final Mission score shall be ranked higher.

9.2. For Overall Awards, including Gold, Silver, and Bronze, the team with the higher Final Mission score shall be ranked higher.

9.3. If the tie persists, the team with the lower mission completion time in the Final Mission shall be ranked higher.

9.4. If a tie remains, the Organising Team shall determine the final ranking based on overall technical merit and mission performance.

10. Code of Conduct & Dispute Resolution

Professional Conduct

10.1. All participants, faculty mentors, judges, volunteers and organisers shall conduct themselves professionally and respectfully throughout the competition.

10.2. Participants shall not engage in any behaviour that disrupts the competition, endangers others, damages property, or brings the competition into disrepute.

10.3. Any form of harassment, discrimination, intimidation, misconduct or abusive behaviour shall not be tolerated and may result in disqualification.

Safety Compliance

10.4. Teams shall be solely responsible for ensuring the safety, reliability and airworthiness of their systems.

10.5. Teams shall comply with all safety instructions, inspection requirements and operational guidelines issued by the organisers.

10.6. The organisers may prohibit the operation of any drone, autopilot, communication system or equipment deemed unsafe.

10.7. Failure to comply with safety requirements may result in penalties, termination or disqualification.

Academic Integrity

10.8. Teams shall submit only original work developed by their team members.

10.9. Teams may utilise commercially available components, open-source software, and publicly available resources, provided that such use complies with applicable licences and competition rules.

10.10. Plagiarism, misrepresentation, falsification of technical information, fraudulent claims, or deliberate concealment of facts shall constitute a violation of competition integrity.

10.11. Teams found to have copied, replicated or substantially reused another team's work without authorisation may be disqualified.

Organiser Authority

10.12. The Organising Team shall have the overall authority to create, interpret, modify and amend the competition rules, guidelines, timelines, evaluation criteria, awards and other competition requirements from time to time.

10.13. The Organising Team reserves the right to inspect, evaluate and verify any information, document, software, hardware or component submitted by participating teams.

10.14. The Organising Team shall have the final authority to determine eligibility, impose penalties, approve problem statement participation and declare competition results.

10.15. The Organising Team reserves the right to modify the competition schedule, problem statement requirements, awards, prizes or operational procedures whenever necessary in the interest of safety, fairness or successful conduct of the competition.

Dispute Resolution

10.16. Participants may raise concerns or seek clarification regarding competition-related matters through the designated communication channels established by the organisers.

10.17. The Organising Team shall make reasonable efforts to review and resolve concerns in a fair and timely manner.

10.18. Decisions of the jury, inspectors and Organising Team regarding technical evaluations, inspections, scoring, penalties, rankings and awards shall be final and binding on all participants.

10.19. Participation in the competition shall constitute acceptance of all rules, guidelines and decisions issued by the Organising Team.

NIDAR
NATIONAL INNOVATION CHALLENGE FOR DRONE APPLICATION AND RESEARCH

Track 1 - Drone Innovation

Problem Statement Brief - NIDAR RescueSwarm

Autonomous Multi-Drone Survivor Search & Aid Delivery Challenge

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Scenario

A severe flash flood has affected a semi-urban settlement located near a river basin. Roads are blocked, several houses are inaccessible, and the local rescue team has received unverified reports of stranded people across a large area of the affected zone. **Telecom towers in the affected region are down, mobile connectivity is unavailable, and ground teams cannot rely on external communication networks for real-time coordination.** Ground access is delayed, and the first few minutes of response are critical for locating survivors and delivering basic medical support.

The rescue command centre has identified a **10-hectare search area** where up to **10 survivors** may be present. Since the area is too large for a single drone to cover efficiently, innovation teams must deploy a **collaborative autonomous multi-drone system** that can rapidly search the area, identify survivors, geotag their locations, and deliver a **survivor kit** to each identified survivor.

The goal of the innovation teams is to **find survivors and deliver them a survivor's kit as quickly as possible**, while ensuring that all drones operate through a single mission system, report mission data to a single operator, and complete the mission with minimal human intervention.

1. Overall Objective

- Teams must deploy **two or more autonomous drones** as part of one collaborative mission system.
- The system must scout a **maximum of 10-hectare land area** provided by the organisers.
- The system must identify and geotag **up to 10 survivors (real humans or dummies)** within the area.
- The system must deliver a **survivor's kit / medical parcel** near the identified survivors.
- The mission must be executed in the **shortest possible time to win** while maintaining accuracy, safety, and autonomy.

2. Drone Configuration & Weight Constraints

- Teams may decide the number, size, and configuration of drones, subject to the minimum requirement of deploying **two or more drones**.
- **Use of commercially available market-ready or ready-to-fly complete drone airframes shall not be permitted.**
- The **combined all-up weight of all deployed drones**, including batteries, onboard systems, sensors, communication equipment, payload mechanisms, and any other component, must not exceed **25 kg**.
- The collective weight limit shall apply to the fully deployed drone system, not to each drone individually.
- Any drone system exceeding the collective weight limit shall not be permitted to fly.

- **The weight of each payload shall be 200 grams, and its dimensions shall be 20 x 10 x 5cm (l x b x h).** Each payload shall be a rectangular box.

3. Mission Planning, Execution & Autonomy

- The organisers shall provide a mission file/boundary file for the assigned area during the setup time at the beginning of the Final Mission. **This mission file will not be provided before the setup time.**
- **Teams shall be provided a maximum of 5 minutes for setup, configuration and mission-file loading before launch.**
- Teams must load the mission file into the Mission Planner / Ground Control Station before launch.
- After the mission is initiated, the drones must autonomously:
  - scout the assigned area;
  - coordinate between multiple drones;
  - detect survivors;
  - geotag survivor locations;
  - assign or execute delivery tasks;
  - drop survivor's kits near identified survivors;
  - return to the launch/landing area, **within a maximum flying time of 30 minutes.**
- Any operator action beyond mission file loading, mission start, safety abort, or emergency recall shall be treated as **manual intervention**.
- Any manual waypoint change, separate drone control, flight-path adjustment, payload drop command, survivor tagging input, or mission re-planning during execution shall be considered **manual intervention**.
- No battery, payload, sensor, communication equipment or other component may be swapped, replaced or added after commencement of the mission.

4. Multi-Drone Collaboration & Mission Control

- All drones must operate as part of a coordinated system and not as independently controlled units.
- The system should demonstrate collaborative mission execution, including area division, task allocation, parallel scouting, delivery assignment, and shared mission progress reporting.
- Mission data from all drones must be reported to a **single operator interface**.
- Use of separate Ground Control Stations, or independent control systems for different drones, shall not be permitted.
- The Mission Planner / Ground Control Station must display:
  - loading of the mission file/mission boundary;
  - live location and status of all deployed drones;
  - tagged location of detected survivors;
  - delivery status of survivor's kits by each drone;
  - basic mission progress and completion status.
  - Live video feed display

- Any system input, command, or operator action outside the above-permitted functions shall be considered **manual intervention**, except for a safety abort or emergency recall.

5. Communication & Network Constraint

- The mission shall be designed and executed assuming a no-external-network environment.
- **Teams shall not rely on GSM, LTE, 5G, public Wi-Fi, internet connectivity, cloud-based communication, or any external network for drone operations, mission execution, data transfer, or coordination.**
- All communication between drones and the Mission Planner/Ground Control Station must be handled through the team's local communication link or onboard autonomy system. Optical-fibre cables, wired links, physical tethers or any other flight-connected cable shall not be permitted.
- The system must operate assuming there is no mobile network or internet connectivity.
- Use of any external network-based communication interface during the mission shall be treated as a violation or manual/external intervention, as applicable.

6. Team Deployment & Human Intervention

- A maximum of **two team members** shall be allowed for setting up drones, payloads, communication systems, and associated equipment **during the permitted 5-minute setup period.**
- No more than **one operator** shall supervise the mission through the Ground Control Station.
- No assistance of any kind shall be allowed from any other team member during setup, launch, mission execution, payload delivery, landing, troubleshooting, or recovery.
- Other team members may be present only as observers and shall not provide verbal, physical, digital, or technical assistance during the mission.

7. Launch, Landing & Field Constraints

- All drones must be launched from and land within a fixed **12 ft x 12 ft launch/landing area**.
- The mission shall be conducted outdoors under standard daylight conditions.
- No part of any drone shall cross or remain outside the launch/landing area during launch or landing.
- The drones must operate only within the assigned mission boundary.
- Landing outside the designated launch/landing area shall be treated as a violation, except in case of an emergency landing triggered for safety reasons.

8. Safety & Failsafe Requirements

- Each drone must have return-to-home capability.
- Each drone must include failsafe features for: low battery; loss of command and control link; geofence breach; mission abort; emergency recall.
- The system must allow the operator to safely abort the mission and recall all drones if required.

NIDAR
NATIONAL INNOVATION CHALLENGE FOR DRONE APPLICATION AND RESEARCH

Track 1 - Drone Innovation

Problem Statement Brief - NIDAR AirMouse

Autonomous GPS-Denied Indoor Search, Mapping & Survivor Localisation Challenge

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Scenario

A building has been severely affected by an earthquake / structural collapse. Several internal passages and rooms are unsafe for immediate entry by rescue personnel because of debris, damaged structures, low visibility, and the risk of further collapse. Initial reports suggest that survivors may be trapped inside different rooms, but their exact locations are unknown.

Since GPS signals are unavailable inside the building, rescue teams need an autonomous aerial system that can enter the structure, navigate through enclosed corridors and rooms, identify survivors, generate a **2D map** of the explored indoor area, and tag the precise location of each survivor on the map.

Innovation teams are required to build an autonomous indoor drone system that can enter a covered maze-like building environment from one designated entry point, navigate through multiple passages and room-like sections, locate up to **6 survivors**, generate a **2D map** of the explored area, tag survivor locations on the map, and exit from a designated exit point.

The goal of the innovation teams is to **autonomously navigate an unknown GPS-denied indoor environment, find survivors, and provide rescue teams with a usable 2D map showing the tagged location of each survivor in the shortest possible time**.

1. Overall Objective

- Teams must deploy an autonomous drone capable of operating in a **GPS-denied indoor environment**.
- The drone must enter the indoor maze from a designated **entry & exit point**.
- The drone must navigate through corridors, turns, junctions, and rooms inside the covered arena.
- **The system must detect up to 6 survivors (real humans or dummies)** placed in different rooms and identify the grid coordinate or grid box where each survivor is located.
- **The system must simultaneously generate and continuously display a 2D map of the explored area** in the Mission Planner / Ground Control Station while the drone is flying.
- The system must tag each detected survivor's location on the generated map by showing a marker on the map.
- The Mission Planner / Ground Control Station **must display a live camera feed** from the drone throughout the mission.
- The mission must be completed in the **shortest possible time** while maintaining safe flight, autonomy, mapping accuracy, and localisation accuracy for survivors.

2. Drone Configuration & Weight Constraints

- **Use of commercially available market-ready or ready-to-fly complete drone airframes shall not be permitted.**
- The **combined all-up weight of the deployed drone**, including batteries, onboard systems, sensors, communication equipment, payload mechanisms, and any other component, must not exceed **10 kg**.
- The drone must be equipped with propeller covers or guards that fully protect the propeller operating area.

3. Arena, Grid & Mission Environment

- The mission shall be conducted within a **covered, maze-like indoor arena** representing corridors, rooms, turns, junctions, and enclosed spaces within a damaged building.
- The total maze/arena size shall not exceed **15 m x 15 m**.
- The arena shall be designed on a modular grid structure. **The top of the indoor space will be covered with a net.**
- **Each corridor shall have a uniform clear width of at least 1 metre and a vertical clearance of at least 8 feet to allow safe drone navigation within** the maze.
- The top of the arena shall be covered to simulate an indoor / GPS-denied environment.
- The arena shall include the same designated **entry point** and **exit point**.
- **The standard room size shall be 2 m x 2 m (l x b).**
- **The height clearance in corridors and all rooms at all times shall be 8 feet.**
- Corridor widths and grid dimensions shall remain uniform across the arena.
- The exact internal layout of the maze may not be disclosed to teams before the mission.

4. Mission Planning, Execution & Autonomy

- **Teams shall be provided a maximum setup time of 5 minutes** to position and prepare the drone, Ground Control Station, communication systems and associated equipment before the mission begins.
- The drone must begin the mission from the designated entry point.
- After the mission is initiated, the drone must autonomously:
  - enter the maze;
  - navigate through the indoor environment;
  - identify corridors, turns, junctions, rooms and obstacles;
  - detect survivors;
  - Identify the grid coordinate or grid box in which each detected survivor is located;
  - simultaneously generate and continuously update a 2D map of the explored area during flight;
  - tag each detected survivor within the relevant grid coordinate or grid box on the generated map;
  - proceed toward the designated exit point;
  - Exit the maze safely.

- **The maximum permitted flying time shall be 30 minutes,** measured from take-off until the drone exits the maze or the mission is terminated.
- The system must not rely on GPS / GNSS-based navigation.
- Any manual control input, path correction, waypoint adjustment, survivor tagging input, or operator-assisted navigation during the mission shall be considered **manual intervention**.
- The operator shall not use FPV goggles or any separate piloting, navigation, monitoring, or video-viewing device during the mission. All authorised mission supervision and live video viewing must take place only through the designated Mission Planner / Ground Control Station.
- The operator may only start the mission and trigger the safety abort/emergency stop if required.
- All mapping, survivor localisation, data processing and transfer of mission data to the Mission Planner / Ground Control Station must occur during flight. **No additional time shall be provided after flying** for generating or completing the map, processing mission data, transferring data, or updating the Ground Control Station.

5. Indoor Mapping & Mission Planner Requirements

- The Mission Planner / Ground Control Station must display:
  - live mission status of the drone;
  - live camera feed from the drone;
  - a 2D map generated and continuously updated while the drone is flying;
  - identified corridors, rooms or sections, wherever technically feasible;
  - the grid coordinate or grid box containing each detected survivor;
  - tagged locations of detected survivors;
  - drone position or estimated drone position within the mapped area;
  - mission progress and completion status.
- The 2D map, survivor locations and mission data must be generated, displayed and transferred to the Mission Planner / Ground Control Station during flight. No post-flight time shall be provided for completing the map, transferring data or making any corrections.
- The displayed map should be usable by rescue teams to understand the approximate layout of the indoor environment and proceed toward the tagged survivor locations.
- Any manual drawing of the map, manual tagging of survivors, or operator-assisted correction of survivor locations during the mission shall be considered **manual intervention**.

6. Survivor Detection & Localisation

- Up to 6 survivors shall be placed at different locations within the maze / room-like sections.
- The drone must detect survivors using onboard sensing and processing.
- The system must identify and tag the grid coordinate or grid box in which each detected survivor is located on the generated 2D map.
- Survivor locations must be shown by clearly identifying the relevant grid coordinate or grid box to guide rescue teams to the corresponding room or section.

- Survivor detection, grid localisation and tagging must occur autonomously and be displayed on the continuously generated 2D map during flight.

7. Communication & Network Constraint

- The mission shall be designed and executed assuming a no-external-network environment.
- **Teams shall not rely on GSM, LTE, 5G, public Wi-Fi, internet connectivity, cloud-based communication, or any external network for drone operations, mission execution, data transfer, or coordination.**
- All communication between drones and the Mission Planner/Ground Control Station must be handled through the team's local communication link or onboard autonomy system.
- The system must operate assuming there is no mobile network or internet connectivity.
- Use of any external network-based communication interface during the mission shall be treated as a violation or manual/external intervention, as applicable.

8. Team Deployment & Human Intervention

- **A maximum of two team members shall be allowed to set up the drone**, Ground Control Station, communication systems and associated equipment **during the permitted 5-minute setup period.**
- No more than **one operator** shall supervise the mission through the Ground Control Station.
- No assistance of any kind shall be allowed from any other team member during setup, launch, mission execution, mapping, survivor detection, exit, landing, troubleshooting, or recovery.
- Other team members may be present only as observers and shall not provide verbal, physical, digital, or technical assistance during the mission.

9. Launch, Landing & Field Constraints

- The drone must be launched from within a fixed **2 feet x 2 feet launch area**.
- No part of the drone shall cross or remain outside the designated launch area before mission start.
- The drone must be sized and configured to operate safely within corridors having a minimum clear width of 1 metre and a minimum vertical clearance of 8 feet.
- The drone must be capable of safely navigating confined indoor spaces without touching walls, ceiling, arena panels, or obstacles.

10. Safety & Failsafe Requirements

- The drone must include emergency stop/mission abort capability.
- The drone must include failsafe features for: low battery; loss of command and control link; geofence breach; mission abort; emergency recall.
- The system must allow the operator to safely abort the mission if required.
- The drone must be designed for safe operation inside confined indoor spaces.

NIDAR
NATIONAL INNOVATION CHALLENGE FOR DRONE APPLICATION AND RESEARCH

Track 2 - Component Innovation

Problem Statement Brief - NIDAR VEGAPilot

Indigenous Processor-Based Flight Controller & Autopilot Challenge

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Scenario

India's drone ecosystem is growing rapidly across defence, agriculture, logistics, infrastructure, disaster response, surveillance, and public service delivery. However, one of the most critical subsystems in any drone is the **flight controller/autopilot**, which serves as the brain of the drone and controls flight stability, navigation, communication, sensor integration, and mission execution.

A large part of India's drone electronics ecosystem still depends on imported flight controllers, processors, microcontrollers, ICs and supporting electronic components. For India to build a truly indigenous drone ecosystem, students and innovators must be able to design, develop, integrate, and demonstrate reliable autopilot systems built around Indian processor platforms and non-Chinese electronic components.

Innovation teams are required to build a **functional drone autopilot system using ARIES v3 Development Boards, based on the indigenous VEGA processor**. Each selected team shall be provided with **two ARIES v3 Development Boards** for development, testing and final integration. Teams must design and build the autopilot hardware and software stack around the ARIES v3 - VEGA Processor Board and demonstrate its ability to control a standard multirotor drone platform.

During the final demonstration, teams shall showcase a demonstration of a drone built from a standard set of drone components sourced from Indian component companies as issued under this RuleBook. These may include an airframe, motors, ESCs, a GNSS module, a battery, a communication module, and other organiser-specified accessories. Teams must integrate their VEGA-based autopilot with these standard components and demonstrate manually piloted mission execution through a Ground Control Station.

The goal of the innovation teams is to **develop an indigenous autopilot that can stabilise, control, navigate, and execute a standard drone mission on a real airframe, while demonstrating reliable integration with sensors, communication systems, and mission-planning** software.

1. Overall Objective

- Teams must build a **functional drone autopilot/flight controller** using the ARIES v3 - VEGA Processor Board as the core processing platform.
- Teams must develop the autopilot hardware and software stack using the ARIES v3 Development Board and only the components included in the organiser-prescribed component list. No additional component may be used to perform or assist any autopilot-related function.
- The autopilot must be capable of controlling a standard multirotor drone platform.
- Each selected team shall be provided with **two** ARIES v3 - VEGA Processor Boards for development and final integration.

- During the final demonstration, teams shall be provided with a list of standard set of drone components sourced from **Indian component companies** for integration and demonstration, which may include:
  - standard airframe;
  - standard motors;
  - standard ESCs;
  - standard GNSS module;
  - standard battery;
  - standard communication module;
  - other standard integration accessories as specified by the organisers.
- Teams must integrate their ARIES v3 - VEGA Processor Board based autopilots with the standard components sourced from Indian component companies as issued under the RuleBook.
- The final demonstration must be performed on the standard airframe as prescribed by the organisers.
- The autopilot must demonstrate stable flight, mission execution, communication with Mission Planner / Ground Control Station, and basic flight safety features.
- Teams must procure their own drone with the exact specified components issued under the RuleBook.
- Teams may leverage open-source ecosystems and protocols such as ArduPilot, PX4, and MAVLink, etc.

2. Application Shortlisting & Processor Application

- Teams must submit an online application describing their proposed approach for building an **ARIES v3 - VEGA Processor Board-based autopilot**.
- The application must include the team's proposed hardware architecture, software/firmware approach, integration and testing approach using the prescribed components and expected demonstration capability.
- All teams shall be invited to deliver an **online technical presentation** before the organising team/jury.
- The online presentation shall evaluate the team's understanding of:
  - ARIES v3 - VEGA Processor Board integration;
  - autopilot hardware architecture;
  - sensor and communication interfaces;
  - flight-control software/firmware approach;
  - indigenous component strategy;
  - safety, testing and demonstration plan.
- Teams shall be scored based on their application and online presentation.
- The **top 100 teams** selected through this evaluation process shall be provided with **two** ARIES v3 - VEGA Processor Boards each for development, testing and final integration and mission.
- Only teams selected in the top 100 shall proceed to the development and final demonstration stage of the challenge.

3. Indigenous Component & Supply Chain Requirements

- The autopilot must be built around the ARIES v3 - VEGA Processor Board as the primary processing unit.
- The complete autopilot must be developed using only the ARIES v3 Development Board and the components included in the prescribed component list issued by the organisers.
- Teams shall not use any additional component to perform, supplement, accelerate, bypass or provide redundancy for flight stabilisation, sensor processing, state estimation, navigation, control calculations, motor-output generation, communication, failsafe logic or any other autopilot-related function.
- Custom PCBs, wiring harnesses, connectors, and mechanical mounting arrangements may be developed only to integrate the prescribed components. They shall not incorporate any unlisted active electronic component or provide an independent processing or flight-control capability.
- No component used inside the fully built autopilot shall be of **Chinese origin**.
- Teams can buy components from the provided links under the Rulebook.
- This restriction shall apply to all components used in the autopilot, including but not limited to resistors, capacitors, inductors, diodes, transistors, voltage regulators, power management ICs, memory ICs, communication ICs, sensor ICs, connectors, PCB assemblies, and any other electronic or electromechanical component used in the autopilot.
- Teams must submit a **Bill of Materials** that clearly identifies the make, model, manufacturer, country of origin, and supplier for each component used in the autopilot.
- Teams must submit supporting documents, declarations, datasheets, invoices or supplier confirmations, wherever required by the organisers.
- Any use of Chinese-origin components in the autopilot shall result in immediate disqualification as decided by the organisers.

4. Autopilot Functional Outcomes

The autopilot must be capable of performing the following functions:

- Read and process inputs from onboard sensors required for flight control.
- Interface with the GNSS module for outdoor positioning and navigation.
- Interface with ESCs and motors to control the multirotor platform.
- Maintain stable attitude control during flight.
- Support basic flight modes, including:
  - arm / disarm;
  - manual / assisted stabilised mode;
  - position hold;
  - altitude hold;
  - autonomous mission mode;
  - return-to-home / return-to-launch.

- Execute a pre-loaded autonomous mission through waypoints.
- Communicate telemetry data to the Mission Planner / Ground Control Station.
- Receive mission commands from the Mission Planner / Ground Control Station.
- Display live vehicle status, location, flight mode, battery status and mission progress on the Mission Planner / Ground Control Station.
- Trigger failsafe actions in case of low battery, communication loss, GNSS issue, or mission abort.
- Log basic flight and mission data for post-flight review.

5. Hardware Design & Integration Requirements

- Teams must develop the autopilot hardware and software stack using the ARIES v3 Development Board and only the components included in the organiser-prescribed component list. No additional component may be used to perform or assist any autopilot-related function.
- The ARIES v3 Development Board shall serve as the sole processing platform for all autopilot-related functions. Teams shall use only the components specified in the organiser-provided component list and shall not add any other processor, microcontroller, flight controller, companion computer, sensor, navigation module, control board or electronic component for performing or assisting such functions.
- The autopilot must include the prescribed components and the supporting integration circuits necessary for power management, sensor interfaces, communication and motor-control outputs. All such circuits and components must conform to the organiser-issued component list.
- Teams may design the required PCB, connectors, power lines, signal lines and interface circuits for integrating the prescribed components. **No additional processor, microcontroller, programmable control device, sensor, flight controller or companion computer may be incorporated into such hardware.**
- The autopilot must be capable of receiving inputs from standard onboard modules and transmitting control outputs to the ESCs.
- During the final demonstration, the autopilot must be mechanically and electrically integrated with the standard components provided by the organisers.
- Teams must ensure proper electrical isolation, voltage regulation, signal integrity and vibration tolerance.
- The autopilot must be packaged or mounted safely on the standard airframe during the demonstration.
- Teams shall not modify or alter any standard components prescribed by the organisers.

6. Software, Firmware & Mission Planner Integration

- Teams must develop or adapt the firmware required to operate the autopilot.
- The software stack must support sensor reading, flight control, navigation, communication, telemetry and mission execution.

- The autopilot must integrate with a Mission Planner / Ground Control Station for:
  - vehicle connection;
  - telemetry display;
  - mission upload;
  - waypoint configuration;
  - flight mode display;
  - battery and health monitoring;
  - mission execution status;
  - return-to-home / mission abort command.

- The system must be capable of executing a standard autonomous mission uploaded via the Mission Planner/Ground Control Station.

- Teams must not rely on simulation-only outputs for final evaluation. The autopilot must demonstrate real hardware integration and flight capability.

7. Final Mission Requirements

- Teams must demonstrate the functional capability of the autopilot by integrating the ARIES V3 Development Board with the standard drone components, sensors, interfaces and communication equipment specified by the organisers.

- The final mission shall evaluate whether the team-developed autopilot can process onboard sensor data, receive operator commands, stabilise a multirotor aircraft, generate reliable motor control outputs, and support safe and controlled flight.

- The final mission shall include:
  - receiving and integrating the standard demonstration components provided by the organisers;
  - integrating the ARIES V3 Development Board with the prescribed IMU, receiver, ESCs, power system and other required interfaces;
  - completing pre-flight checks and demonstrating correct sensor readings, motor outputs and communication between the integrated system components;
  - safely arming and disarming the drone;
  - taking off from a designated launch area under operator command;
  - maintaining a stable hover for a specified duration;
  - demonstrating controlled response to roll, pitch, yaw and throttle inputs;
  - flying forward, backwards, and laterally while maintaining flight stability;
  - executing controlled changes in heading;
  - navigating through or around a predefined course consisting of gates, markers and obstacles;
  - passing through the gates or around the obstacles in the prescribed sequence;
  - maintaining stability while transitioning between different directions of movement;
  - stopping and maintaining a stable hover at designated points along the course;
  - demonstrating controlled recovery after the operator returns the directional control inputs to neutral;
  - navigating to a designated landing area under operator command;
  - landing safely within the prescribed landing zone;
  - recording relevant flight-control data, including sensor inputs, operator commands and motor outputs, for post-flight review; and
  - making the autopilot hardware, software, integration details and flight logs available to the organisers for evaluation

- Manual flight commands shall not be permitted during the autonomous mission. All sensor processing, flight stabilisation, control calculations and motor-control outputs must be performed by the autopilot developed using the ARIES V3 Development Board.

- Direct control of individual motors, bypassing the team-developed autopilot, or using any separate flight controller, processor, microcontroller, companion computer, sensor-processing unit or other unlisted component to perform or assist an autopilot-related function shall not be permitted and will result in immediate disqualification.

- The obstacle course shall be designed to evaluate the stability, controllability, command response and integration of the autopilot rather than the piloting speed of the operator. The organisers may prescribe the dimensions, sequence, operating height, hover duration, landing tolerance and other mission parameters before the final demonstration.

- The final mission must establish that the system is not merely a bench-level prototype, but a functional student-developed autopilot capable of stabilising and controlling a standard multirotor platform during a manually commanded flight.

8. Communication Requirements

- The autopilot must communicate with the Mission Planner/Ground Control Station via the standard communication module provided or approved by the organisers.

- The system must relay key flight telemetry, including:
  - drone location;
  - altitude;
  - flight mode;
  - battery status;
  - mission status;
  - failsafe status;
  - communication link status.

- The Mission Planner / Ground Control Station must allow the operator to monitor mission data during the demonstration.

9. Team Deployment & Human Intervention

- A maximum of **two team members** shall be allowed for setting up drones, payloads, communication systems, and associated equipment.

- No more than **one operator** shall supervise the mission through the Ground Control Station.

- No assistance of any kind shall be allowed from any other team member during setup, arming, launch, mission execution, landing, troubleshooting, or recovery.

- Other team members may be present only as observers and shall not provide verbal, physical, digital, or technical assistance during the mission.

10. Safety & Failsafe Requirements

- The autopilot must support safe arming and disarming.
- The autopilot must include failsafe features for:
    - low battery;
    - loss of command and control link;
    - GNSS signal issue;
    - unstable flight condition;
    - mission abort;
    - return-to-home / return-to-launch;
    - emergency landing.
- The system must allow the operator to safely abort the mission if required.
- The drone must not be flown unless the organisers are satisfied with the safety of the autopilot, the integration, the communication link, and the mission setup.
- Any unsafe behaviour during pre-flight inspection or demo flight may result in mission termination, penalties or disqualification.

11. Standard List of Drone Components for Demonstration

Shortlisted teams shall be provided with a list of a standard set of drone components sourced from **Indian drone component companies**. Teams must integrate their autopilot with these components and demonstrate the required mission outcomes.

The component list issued by the organisers shall be exhaustive for all autopilot-related hardware. Teams may use different makes, models or specifications only where expressly permitted in the list or approved in writing by the organisers before integration.

An unlisted component found to be performing or assisting an autopilot-related function may result in rejection of the system or disqualification.

The standard demonstration kit may include:

No.	Component	Make / Model / Specifications
To be announced soon.		

ABOUT DFI

The Drone Federation India (DFI) is a non-profit industry body pioneering the vision of making India a global drone manufacturing and export hub. We serve as an apex body that represents over 550+ drone companies and 5500+ drone professionals across the entire drone ecosystem.

We promote ease of doing business for drone companies, widespread adoption of drone technology across all sectors of the economy and build a skilled community of drone professionals to deliver high-quality world-class products from India.

We bring together government, industry, academia, experts and civil society to build a safe and secure unmanned aerial ecosystem in India.

ABOUT SWAYAAN

The Ministry of Electronics and Information Technology (MeitY) has launched 'SwaYaan – Capacity Building for Human Resource Development in Unmanned Aircraft Systems' in September 2022 with the goal of fostering a national industry-academia collaborative ecosystem for Drone/UAS & related technologies.

The initiative aims to train as many professionals in the UAS sector by 2027 through 1,500+ activities nationwide encompassing five major technical areas towards making India the Global Drone Hub by 2030.

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